

Roll No.

TCE-701

**B. TECH. (CIVIL) (SEVENTH SEMESTER)
MID SEMESTER EXAMINATION, Oct., 2022**

DESIGN OF STEEL STRUCTURES

Time : 1½ Hours

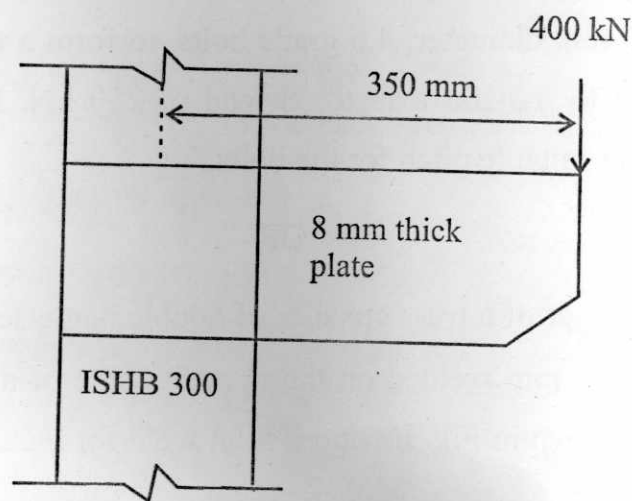
Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.
(ii) Each sub-question carries 10 marks.

1. (a) Enlist some of the advantages and disadvantages of using a steel structure over RCC structure. What parameters a designer must keep in mind before designing a steel structure ? 10 Marks (CO1)

OR

- (b) A bracket is bolted to the flange of a column as shown below using 8 mm thick plate. Using M20 bolts of grade 4.6. Design the connection for $f_u = 410$ MPa, $f_{ub} = 400$ MPa, $f_y = 250$ MPa, $f_{yb} = 240$ MPa. 10 Marks (CO1)



All dimensions are in mm

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2. (a) You have learnt about the tensile strength of mild steel, so can you explain the engineering stress strain curve for mild steel in detail.

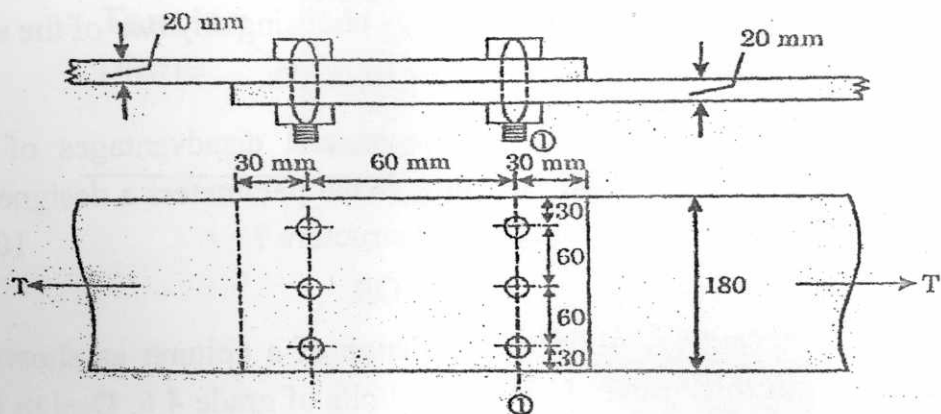
10 Marks (CO1)

OR

- (b) Find the efficiency of the lap joint shown below.

Given: M 20 Bolts of grade 4.6 are used and plate is Fe 410 with f_y 250 MPa (Shear plane cuts the thread).

10 Marks (CO1)



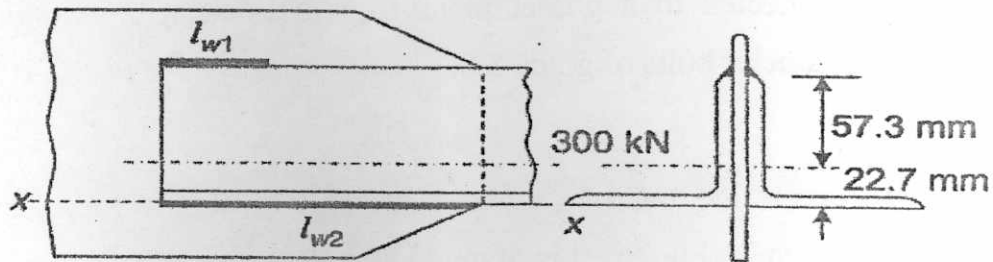
3. (a) Two flats (Fe 410 grade steel), each 210 mm $\times$ 8 mm, are to be jointed using 20 mm diameter, 4.6 grade bolts, to form a lap joint. The joint is supposed to transfer a factored load of 250 kN. Design the joint and determine suitable pitch for the bolts.

10 Marks (CO1)

OR

- (b) A tie member of a truss consists of double angle section, each 80 mm $\times$ 80 mm $\times$ 8 mm welded on the opposite side of a 12 mm thick gusset plate as shown in Fig. Design a fillet weld for making the connection in

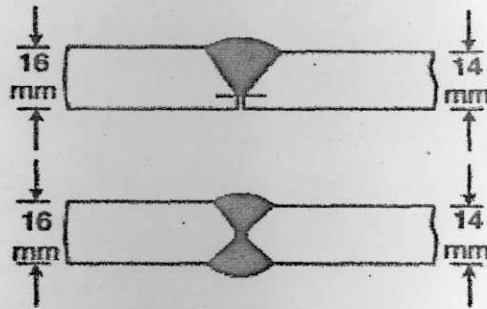
the workshop. The factored tensile force in the member is 300 kN. (Fe 410 grade steel for both connection and members). 10 Marks (CO1)



4. (a) Draw and explain various types of welding connections. Explain size of weld, throat of weld, and length of weld. 10 Marks (CO2)

OR

- (b) Two plates of 16 mm and 14 mm thickness are to be joined by a groove weld as shown in Fig. 1. The joint is subjected to a factored tensile force of 430 kN. Due to some reasons the effective length of the weld that could be provided was 175 mm only. 10 Marks (CO2)



Check the safety of the joint if :

- (i) Single-V groove weld is provided
- (ii) Double-V groove weld is provided

Assume the plates to be shop welded.